Philosophy of Space and Time

Lecture 1: *Introduction and Zeno's Paradoxes*

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Department of Philosophy, UCLA · March 31, 2026

Today's Plan

1. Course Overview
2. Zeno's Paradoxes
3. Initial Responses
   - Sums of Infinite Series
4. Responses to Paradoxes of Composition
   - Composition of the Continuum
   - The At-At Theory of Motion

Welcome

?? *What is the nature of space and time?*
How do we justify answers to this question?
How have the answers changed, and why?

These questions sound simple, but answering them draws on philosophy, mathematics, and physics in surprising ways.

Course Structure

The course is organized in four parts:

1. **Motion and Geometry** Weeks 1–2
   Zeno's paradoxes; Euclidean and non-Euclidean geometry; Conventionalism
2. **17th-Century Debates** Weeks 3–4
   Descartes; Newton's account of motion; Leibniz and relationalism
3. **Modern Spacetime Theories** Weeks 5–8
   From space to spacetime, to dynamical spacetime
4. **It's About Time** Weeks 9–10
   The order of time in contmporary physics

Required Texts

1. Tim Maudlin, *Philosophy of Physics: Space and Time*
2. Robert Geroch, *General Relativity from A to B*
3. Carlo Rovelli, *The Order of Time*
4. *Online Readings* posted to the course website

Available through Bruin Digital (see syllabus for details).

Evaluation

- **Attendance & Participation** 10%
- **Short Assignments** (weekly reading questions) 15%
- **Two In-Class Exams** (May 5, May 28) 40%
- **Final Exam** (June 11) 35%

No particular background in mathematics or physics assumed. The course is self-contained—but an aptitude for careful, abstract reasoning is essential.

Two Themes

Structural Questions

What are the *properties* of space and time?

- Finite or infinite? Continuous or discrete?
- What geometry?
- Contrasts between space and time?
- In what sense do space and time exist?

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Epistemological Questions

How do we *know* anything about the structural properties of space and time? What *kind* of justification can we give?

Ideas about space and time have always been **tied to theories of motion**:

- Motion = change of position. But what do we need to assume to define this?
- Precise characterization of motion requires claims about the structure of space and time.

Space, Time, and Motion

Ideas about space and time have always been **tied to theories of motion**:

- Motion = change of position. But what do we need to assume to define this?
- Precise characterization of motion requires claims about the structure of space and time.
- Our understanding of both has changed dramatically, driven by developments in mathematics and physics.

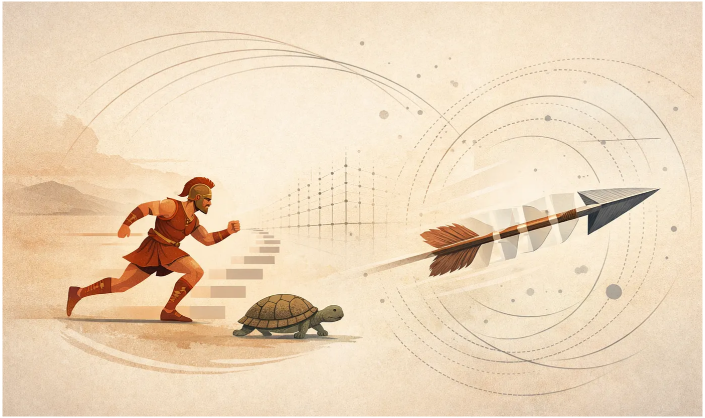
Space, Time, and Motion

We begin with the oldest and most fundamental challenge:

?? *Is continuous motion even coherent? Zeno of Elea argued that our basic assumptions about space, time, and motion lead to outright contradictions.*

Zeno’s Paradoxes

Zeno of Elea (c. 490–430 BCE)



(Image credit: Medium.com)

- Ordinary ideas about space, time, and motion lead to **contradictions**

Four Paradoxes

Paradoxes of Traversal

1. **The Dichotomy**
2. **Achilles & the Tortoise**

Can a journey composed of infinitely many sub-journeys ever be completed?

Paradoxes of Composition

3. **The Arrow**
4. **Infinite Divisibility**

How is time composed of instants? How is a line composed of points?

Paradox 1: The Dichotomy

The Argument

P1: To traverse interval AB, one must traverse all subintervals $\frac{AB}{2^n}$ $(n = 1, 2, 3, \ldots)$.

Two formulations:

- **Progressive**: Achilles can never *reach* the finish line.
- **Regressive**: Achilles can never even *start*!

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CNC: Therefore it is impossible to traverse AB.

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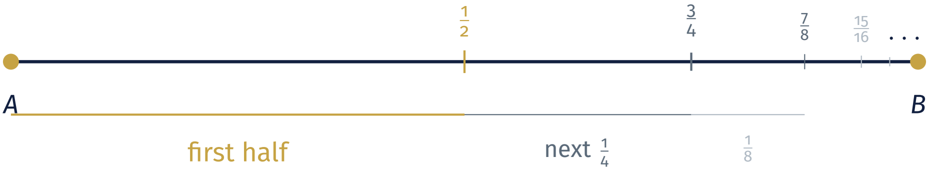
The Dichotomy — Visualized



Before reaching B, Achilles must reach $\frac{1}{2}$, then $\frac{3}{4}$, then $\frac{7}{8}$, … Can he complete *infinitely many* tasks in finite time?

More Carefully...

Consider Achilles running a 100 m dash.

P1: Any distance can be divided in half.

P2: All segments have finite length.

P3: Length of a segment = sum of its parts.

Assumptions about space

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P5: An infinite sum of finite times is infinite.

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C2: (from C1, P3): Achilles traverses an infinite sum of finite segments.

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C1: (from P1, P2): Any segment is divided into infinitely many sub-segments.
C2: (from C1, P3): Achilles traverses an infinite sum of finite segments.
C3: (from C2, P4, P5): Achilles requires *infinite time*.

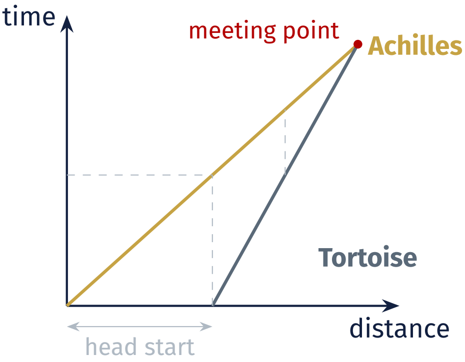
Which premise should we challenge?

Paradox 2: Achilles and the Tortoise

The tortoise has a 100 m head start. When Achilles reaches 100 m, the tortoise is at 110 m. When he reaches 110 m, the tortoise is at 111 m. And so on . . .

Conclusion: Achilles *never* catches the tortoise.

(Structurally similar to the Dichotomy: the “catching point” is the limit of a series.)



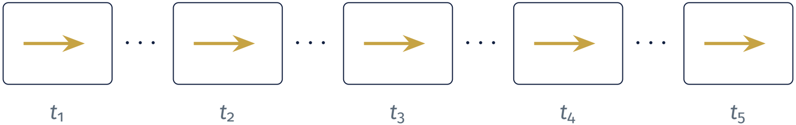
Paradox 3: The Arrow

The Argument

A-P1: At each and every instant, the arrow is at rest.

A-P2: Any interval of time is composed of instants.

A-CNC: Therefore, during any interval, the arrow is not moving.



At each instant the arrow occupies a fixed position—how can it be *moving*?

Argument for A-P1

?? *Why think the arrow is "at rest" at each instant?*

- An instant has *no parts*.
- Motion $=$ being at different positions at different times.
- Within a single instant there is no room for change of position.
- Therefore: at any given instant, the arrow is not moving.

Paradox 4: Infinite Divisibility

The Argument

ID-P1: A line segment is composed of infinitely many points.

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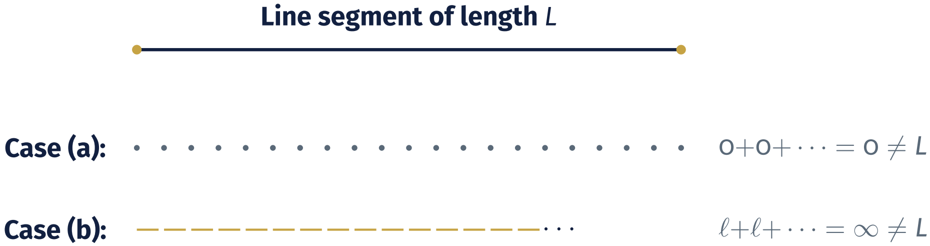
ID-P4: Each point has either (a) zero magnitude, or (b) some finite magnitude (say, ℓ).

Dilemma: both options in ID-P4 contradict ID-P3:

- If (a): $0 + 0 + 0 + \cdots = 0$. But the segment has positive length!
- If (b): $\ell + \ell + \ell + \cdots = \infty$. But the segment is finite!

Infinite Divisibility — Visualized



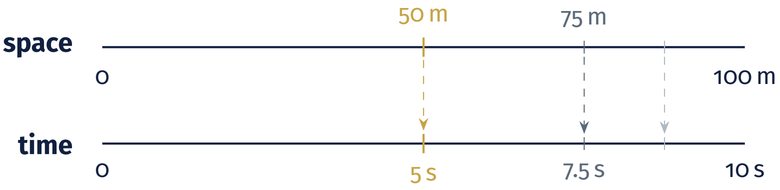
How can a line of finite length be composed of points? This gets at the structure of the **continuum**.

Break

Initial Responses

Aristotle's Response to the Dichotomy

- Aristotle *concedes* that you cannot traverse an infinite *distance* in finite time. But denies that the Dichotomy establishes this!
- Key move: **divide time the same way you divide space**.



Each sub-distance maps to a sub-time. The total time is finite—denying P5.

Actual vs. Potential Infinity

Aristotle also draws a distinction:

- **Actual infinity**: a completed infinite collection that exists all at once.
- **Potential infinity**: the line can always be divided *further*, but the infinite collection is never completed.

Achilles does nothing to “actualize” each midpoint as he runs past it.

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Compare: an obsessive runner who *stops* at each halfway point really would never finish! (That would be “actualizing” the infinity.)

Actual vs. Potential Infinity, 2

But deeper problems remain:

1. What does it mean to "add up" infinitely many quantities?
2. How is a line composed of dimensionless points?

Response 1: Sums of Infinite Series

The critical premise is (generalizing from time to all quantities):

P5: An infinite sum of finite quantities must be infinite.

But **is this true**?

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What does an infinite sum mean? "Finite addition" and "infinite addition" are fundamentally different. Infinite sums not defined adequately until the *19th century*.

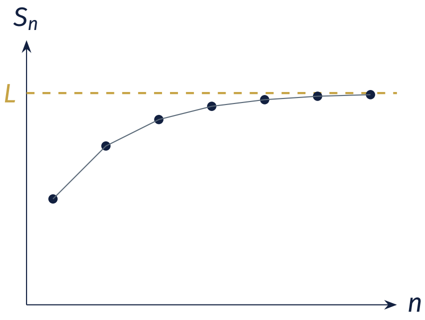
Limits and Convergence

Limit of a sequence: A number L such that the terms get arbitrarily close to L.

Sum of a series: Consider the **partial sums**:

$$S_n = a_1 + a_2 + \cdots + a_n.$$

If $S_1, S_2, S_3, \ldots$ converges to a limit S, then *by definition* the infinite sum equals S.



Partial sums converging to a finite limit

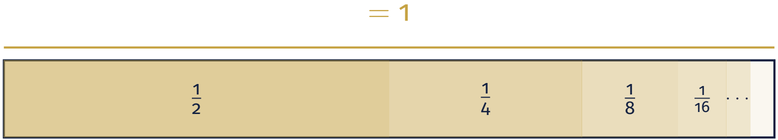
Achilles' Infinite Sum

Suppose Achilles runs 100 m in 10 s (constant speed).

Segment	Distance	Time	Running total
1st	50 m	5 s	5 s
2nd	25 m	2.5 s	7.5 s
3rd	12.5 m	1.25 s	8.75 s
4th	6.25 m	0.625 s	9.375 s
⋮	⋮	⋮	⋮
$\sum$	**100 m**	**10 s**	$\to$ **10 s**

$$5 + 2.5 + 1.25 + 0.625 + \cdots = \sum_{n=1}^{\infty} \frac{10}{2^n} = \mathbf{10\ s}$$

The Geometric Series



$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots = \sum_{n=1}^{\infty} \frac{1}{2^n} = 1$$

The **infinite sum of finite quantities can be finite**.

This directly undermines the key premise of the Dichotomy and the Achilles.

Does This Fully Resolve the Dichotomy?

The mathematical theory shows the *sum* is finite. But a philosopher might press:

?? *Knowing the sum is finite tells us how long the journey takes. But does it explain how infinitely many sub-tasks get completed?*

- Mathematics of convergent series tells us the total time is finite.

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- One response: the "tasks" are only potentially there. Achilles just runs; *we* create "tasks".

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?? *Knowing the sum is finite tells us how long the journey takes. But does it explain how infinitely many sub-tasks get completed?*

- Mathematics of convergent series tells us the total time is finite.
- *Philosophical* question about completing infinitely many tasks persists?
- One response: the "tasks" are only potentially there. Achilles just runs; *we* create "tasks".
- And convergent series don't help with the *other* two paradoxes ...

Four Paradoxes

Paradoxes of Traversal

1. The Dichotomy
2. Achilles & the Tortoise

Can a journey composed of infinitely many sub-journeys ever be completed?

Paradoxes of Composition

3. The Arrow
4. Infinite Divisibility

How is time composed of instants? How is a line composed of points?

Brief Recap

We saw that **convergent series** (arguably) resolves Dichotomy and Achilles: an infinite sum of finite quantities *can* be finite.

Limit of a sequence: A number L such that the terms get arbitrarily close to L.

Sum of a series: Consider the **partial sums**:

$$S_n = a_1 + a_2 + \cdots + a_n.$$

If $S_1, S_2, S_3, \ldots$ converges to a limit S, then *by definition* the infinite sum equals S.

(Note: not all series converge! Claim: series relevant to dichotomy does converge.)

Two paradoxes remain unanswered:

- **The Arrow**: if the arrow is motionless at every instant, when does it move?
- **Infinite Divisibility**: how can dimensionless points compose a line of finite length?

(And even for the Dichotomy, we asked whether the mathematics really settles the philosophical question.)

Recall: Infinite Divisibility Paradox

The Argument

ID-P1: A line segment is composed of infinitely many points.

ID-P2: The length of a segment equals the sum of its parts.

ID-P3: All line segments have a definite, finite length.

ID-P4: Each point has either (a) zero magnitude, or (b) some finite magnitude (say, ℓ).

Dilemma: both options in ID-P4 contradict ID-P3:

- If (a): $0 + 0 + 0 + \cdots = 0$.
- If (b): $\ell + \ell + \ell + \cdots = \infty$.

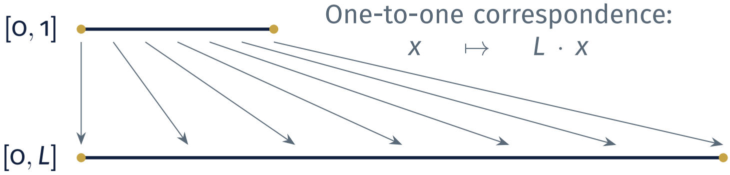
The Continuum Problem

The convergent-series answer does *not* resolve Infinite Divisibility:

- There the sum is $\ell + \ell + \ell + \cdots$ (equal terms) — this does *not converge*.
- The puzzle is about the *composition* of the line, not about subdividing a journey.

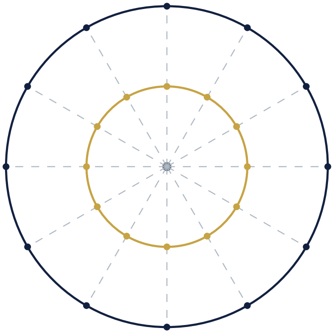
?? *How can a collection of dimensionless points constitute a line segment with definite, finite length?*

All Segments Have the Same "Number" of Points



Both segments contain the **same "number"** of points—regardless of length!

Concentric Circles



Each radial line pairs a point on the inner circle with a point on the outer.

Same “number” of points—but different circumferences!

Length $\neq$ counting points.

Countable vs. Uncountable Infinity

- **Countably infinite**: elements can be listed $a_1, a_2, a_3, \ldots$ (one-to-one correspondence with $\mathbb{N}$). Examples: natural numbers; rationals.
- **Uncountably infinite**: Cantor proved (1874) that the real numbers *cannot* be listed this way.

The points on a line segment form an *uncountable* set.

Our definition of "infinite sum" only applies to **countable** sums — so the idea of summing up point-lengths to get the length of the line *does not apply*.

Modern Resolution

The Metric

The **length** (or *measure*) of a line segment is *not* determined by summing up the sizes of its constituent points.

- The set of points gives you the *topology* (which points are near which), but not the *metric* (how far apart they are).
- A **metric** is additional structure imposed *on top of* the point set.
- This is why segments with the “same number” of points can have different lengths.

Zeno assumes that length must be built from the lengths of points — the modern view rejects this.

Recall: The Arrow Paradox

The Argument

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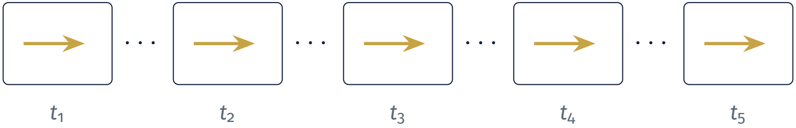
A-P2: Any interval of time is composed of instants.
A-CNC: Therefore, during any interval, the arrow is not moving.



Response to the Arrow: The At-At Theory

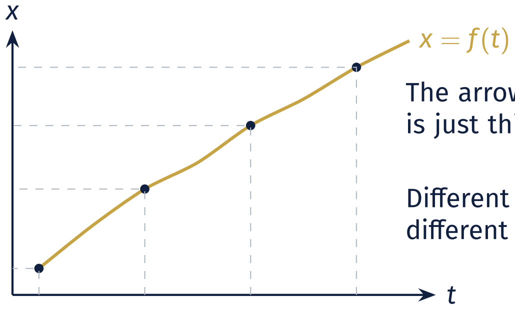
- Aristotle's response: deny A-P2, time is not composed of indivisible instants.
- A better move, first articulated by **Bertrand Russell**:

The At-At Theory

To be "in motion" is simply to be *at* the appropriate position *at* each moment of time. Motion is captured by a **function** $f : t \mapsto x(t)$.

The arrow is "moving" just in case f assigns it *different positions at different times*. No "process" at an individual instant is needed.

The At-At Theory — Visualized



The arrow's *motion*
is just this curve.

Different positions at
different times $=$ moving.

Instantaneous Velocity

?? *If the arrow doesn't "move" at a single instant, can we still say it has a velocity at that instant?*

Yes—the calculus provides a rigorous definition:

$$v(t) = \lim_{\Delta t \to 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} = f'(t)$$

- Instantaneous velocity = **derivative** of the position function.
- Defined via a limit involving *nearby* instants—not by what happens "during" a single instant.
- Even granting A-P1, an arrow can have a well-defined nonzero velocity at each instant.

Challenging A-P1

A deeper objection: **deny A-P1 outright**.

- A-P1 says the arrow is "at rest" at each instant.
- But perhaps rest *simply doesn't apply* to a single instant.
- Crucial difference between:
  - "the arrow has zero velocity" (a substantive claim), and
  - "the question of motion is inapplicable at a single instant."
- The at-at theory says motion is defined *over intervals*; asking whether something moves "at" an instant is a **category error**.

For Next Time

- **Read:** Huggett, *Space from Zeno to Einstein*, Ch. 3 (online as C 1)

On Thursday we continue with:

- Recap of the **composition of the continuum** and the at-at theory of motion
- Next topic: **Euclidean and non-Euclidean geometry**

Questions?

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